

Smart Warehouse System: General Description

In the course, we consider an autonomous smart warehouse system and focus on studying its different aspects and components in different assignments.

Some inspirational pictures depicting the system can be found at the end of the description

High-level System Description

Smart warehouse is a fully automated storage and shipment facility. It is equipped with the autonomous robots that can transport labelled goods (called boxes for simplicity) between the multi-level shelves and collection points.

We assume that there are 10 robots operate in the warehouse. The plan of the warehouse looks as shown in Fig. 1.

Blue rectangles are shelves

Green rectangles are collection points

Red arrows are robotic arm capable of placing the boxes on the conveyer belt and removing them from the conveyer belt when they arrive at the warehouse (making the box to appear and disappear at the conveyer belt).

We assume that the boxes just appear on the conveyor belt when they arrive to the warehouse and should be transported to the shelves. In the similar way, the boxes disappear from the conveyor belt when they are brought to it for shipment.

Yellow lightings are charging stations for the robots.

Shelves

Each place in the warehouse is unique and identified by three numbers: shelf, level, position. There are 20 shelves, each has 3 levels, and 5 positions. For instance, the box on 2nd shelf, 3rd level, 1 position will have the place (2,3,1)

Boxes

Each box has an unique RFID tag attached to it. When it arrives at the warehouse, the warehouse management system (WMS) assigns it the place at which it should be stored. Correspondingly, in its database WMS keeps track of box-place assignments.

Robots hardware

Each robot has a mobile base, an extendable arm attached to it and a storage space. The robot can move forward, backward and turn to the desired angles. The mobile base accepts the commands to move in a given direction with a given speed. We assume that when the base is commanded to move with the given speed, it immediately gets to that speed, i.e., we neglect acceleration. Similarly, when the robot is commanded to stop, it stops immediately, i.e., we neglect deceleration.

The robot is equipped with an extendable arm, which can pick up a box from a shelf and put it on the robot's storage space. It can also take the box from the robot's base and put it on the shelf. The arm is equipped with the RFID tag reader, i.e., it can check the RFID of the box that it handles. The arm can also detect an absence of the desired RFID tag in its vicinity. The arm can be commanded to extend to the

desired level and activate or deactivate the grip. It can also be commanded to pick up the box from the robot's storage space.

The robot's storage space is equipped with the weight sensor, which allows to detect whether the box is on the robot. The robot's storage space is just a place on the top of the robot where a box can be put.

The robot has a battery and a sensor that detects the level of the battery charge. A robot can query the sensor. There is also a predefined function, which translates the battery charge into the possible travel distance. We assume that the arm manipulation consumes a constant predefined amount of charge.

The robot will stop and cease the communication if the battery level becomes critically low.

Each robot is also equipped with a radar. It allows it to detect obstacles on its way and recognize whether they are moving or static as well as to measure a distance to them.

Each robot is equipped with positioning system, i.e., at any moment of time the robot "knows" its coordinate in the warehouse floor. The positioning system is reliable and accurate.

Robot's communication system:

Each robot has an unique ID known to WMS and other robots. The robot can communicate with the WMS. It receives the orders to bring the box from the corresponding place or fetch it and bring to the collection points. The WMS sends the robot both the place and RFID of the box. The WMS also send the routes to the robots (more about it later).

The robot should also communicate with the WMS when it fails to complete its operation due to some reasons. It also should send a notification to the WMS when it decides to abort its current assignment and move to the charging station.

The robot communicate with WMS and send its coordinates, battery status etc.

The robots can also communicate with each other, e.g., when resolving conflicting directions as well as conveyor belt and charging stations.

The robot has a processor and corresponding software installed. It can perform some not very computationally-intensive operations.

The workflow in the warehouse

The workflow in the warehouse is as follows:

1. Boxes arrive at the collection points of the warehouse, where they are getting loaded on the conveyor belt. This operation is outside of the scope of our consideration
2. The box is registered by WMS. It receives a unique RFID tag (it is getting attached to it) and WMS assigns a place to the box.
3. After this is done, WMS chooses an idle robot to transport the box to the designated place (detailed description of this will be given later)
4. WMS requests the battery status of the robot and either commands the robot to approach the corresponding collection point (this position is the same for all robots) or move to a certain charging station. In the former case, WMS chooses another robot for the assignment.
5. When the robot has approached the collection point, it coordinates with the conveyor belt to collect the box. The conveyor belt stops, let the robot to pick up the box from it and then continues, i.e., the robot's arm places the box from the conveyor to the robot base.

6. If the loading operation is successfully completed then the robot sends the notification to the WMS that it has accepted the assignment to transport the box to the requested place.
7. The WMS analyses the current positions of the robots and (re)-computes the routes of all robots and sends them to them.
8. After the conveyor belt is emptied, a new incoming box is handled in the same way.

Bringing the boxes from the shelves follows the similar script but in the reversed order.

All robots follow the same behavioural pattern when en-route.

Each robot periodically received the route updates from the WMS. The robot (locally) translates it into a set of short-term routes and follows them. While moving the robot periodically checks the radar readings and evaluate safety of its next move. The “vertical” spaces between the shelves is narrow, so it can fit only two robots travelling in the same or opposite directions. The “horizontal” spaces are wide and allow for several robots to travel or perform maneuvers.

When the robot arrives at the required position it checks the correctness and safety of the assigned operation. If correctness and safety are established, it manipulates the its arm to perform the requested box placing or fetching and confirms success to WMS. If the correctness or safety are not confirmed then the robot sends the corresponding diagnostics message to WMS and waits for further commands.

When the robot moves, its arm should be in “retracted parked” position.

Warehouse Management System (WMS):

WMS coordinates the work of the entire warehouse. It has powerful processing capabilities and can run computationally-expensive algorithms and communicate with all robots and equipment in the warehouse. The main goal of the WMS is to keep track of boxes and robots and ensure that each box is eventually put in the right place or fetched from the shelf and delivered outside of the warehouse.

WMS has fairly good, though not entirely complete, observability of the warehouse state. It “knows” the positions of the robots, their assignments etc. Hence, WMS can perform safe and efficient route planning. The goal of the route planning algorithm is to ensure that an optimal balance between the length of the route and safety is achieved for all robots.

WMS also handles exceptional situations, e.g., when a box is dropped outside of robot or other machine. The system should be put in a safe state and a human supervisor resolve the problem, update the state and resume functioning of the system.

However, WMS cannot foresee all possible situations and deviations. Hence, the part of ensuring dependability and mission completion is delegated to the robots. While WMS is responsible for overall robot routing, each robot is responsible for safety of its own operation as well as conflict resolution with other robots.

Assumptions:

WMS has a map of the warehouse with all the shelves on it. WMS knows the positions of the charging stations and can request the number of vacant “plugs”.

You can initialize the warehouse in completely empty but then you will need to perform a number of “loading” operations first to start to experiment with fetching. Alternatively, you can populate WMS record (in whatever form you will decide to program it) with some random box placements.

Initially all robots are fully charged and distributed around charging stations.

Communication is reliable and does not have delays. We do not get deep into the details of how communication is implemented and can assume that WMS can send messages and data to robots and equipment, robots can “talk” to each other and WMS.

It is assumed that the rate of requests to handle boxes is high and hence all available robots should be utilized.

Note:

The description of the system is given at a rather high level. Throughout the course you will develop a good understanding of how dependable systems are developed and can by yourself reconstruct the details. However, if any ambiguities are spotted in the requirements then the corresponding notes and explanations will be posted on the announcement board.

Figure 1. Schematic representation of warehouse

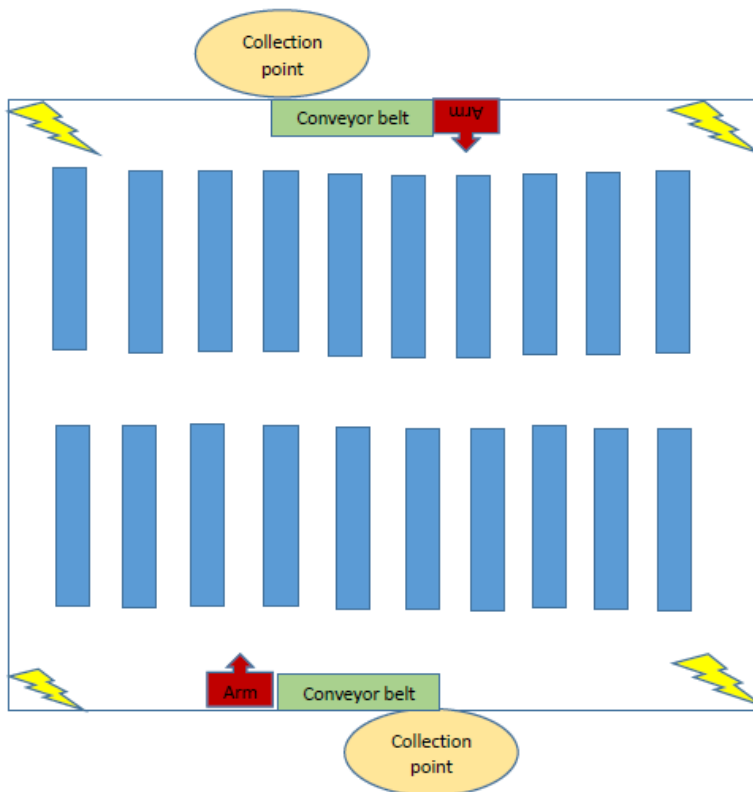


Figure 2. Autonomous robot



Figure 3. Inspirational picture of a smart warehouse

